Survival Impact of Targeted Vaccine in Inflammatory Breast Cancer:

Matched Analysis of Population Registry and Clinical Cohort Data

2026 Biostatistics MS Capstone

UNIVERSITY *of* WASHINGTON

Group Members

- Dr. Ying Liu
  - PhD
  - Cancer Vaccine Institute
- Enya Liu
  - Biostatistics MS Capstone, University of Washington
- Xiao Shi
  - Biostatistics MS Capstone, University of Washington
- Luyi Tong
  - Biostatistics MS Capstone, University of Washington
- Jieyi Xu
  - Biostatistics MS Capstone, University of Washington

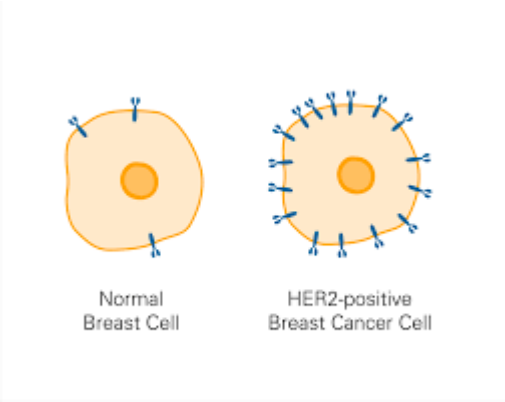
Background

Inflammatory Breast Cancer (IBC) is a rare but highly aggressive subtype of breast cancer

- Represents ~2–4% of cases but ~10% of breast cancer–related deaths
- Typically diagnosed at stage IIIb or higher

Human Epidermal Growth Factor Receptor 2 (HER2) overexpression is more common in IBC

- ~30–40% of IBC vs. ~15–20% of non-IBC
- HER2-targeted therapies (e.g., trastuzumab) have improved outcomes, but long-term survival remains suboptimal



Background

HER2-directed vaccines

- Aim at inducing durable anti-tumor immunity
- Early data suggest potential survival benefit in HER2+ IBC patients
- Because the CVI vaccine cohort has no internal control arm, our study requires an external real-world comparison group.

Background

SEER as an ideal reference cohort

- National, population-based cancer registry
- Includes detailed information on tumor characteristics, treatment, and survival
- Enables benchmarking of vaccinated HER2+ IBC outcomes against standard-of-care populations

Objectives & Hypotheses

Objectives

- Compare overall survival between vaccinated HER2+ IBC patients (CVI) and matched SEER controls.
- Assess whether the association between vaccination and survival varies by hormone receptor status, age, or stage.

Hypotheses

- *Primary:* Vaccination is associated with better overall survival compared with matched SEER controls.
- *Secondary:* The vaccine–survival association may differ across subgroups (HR status, age, stage).

Method | SEER

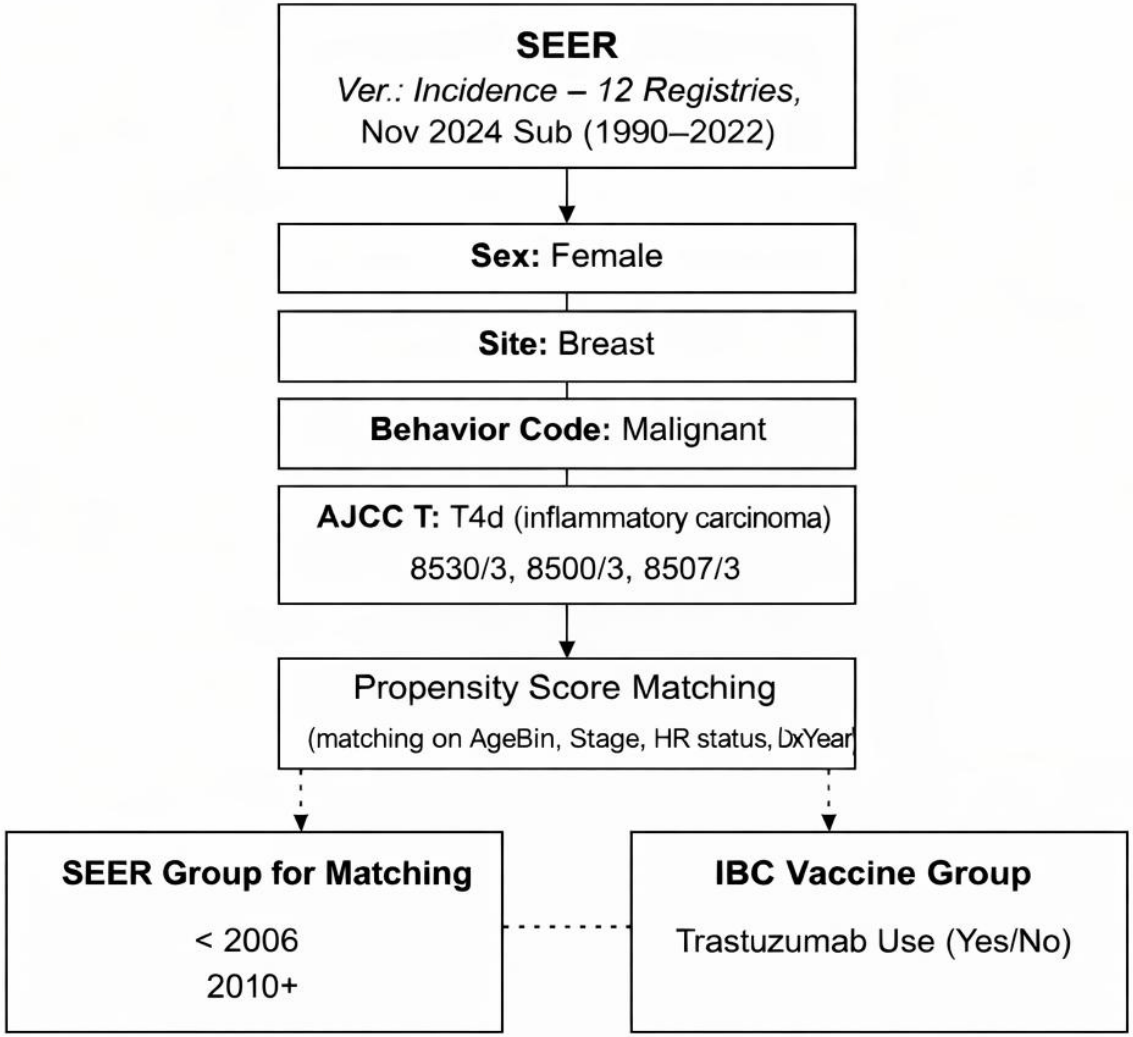


Figure 1. SEER Control Selection Criteria and Workflow

Study Design

Retrospective Matched Cohort Study

- **CVI:** HER2+ IBC patients receiving HER2-directed vaccine
- **SEER:** Unvaccinated HER2+ IBC population (external control)
  - IBC is extremely rare → randomized trials not feasible;
  - CVI lacks internal controls → SEER used as external comparison arm
- Match on **disease features** (IBC/HER2/stage) ; balance **prognostic factors** (age, diagnosis year, HR status)
- Follow-up from **6-month landmark** to death or last contact

Variables

Outcome (Time-to-event): *Overall Survival from diagnosis*

Exposure: *HER2-directed vaccine (Yes for CVI, No for SEER)*

Covariates:

- ***Demographics:*** age, sex, race/ethnicity
- ***Tumor:*** stage, grade, histology, laterality, primary site
- ***Biomarkers:*** hormone receptor (HR) status, HER2 status
- ***Treatment proxies:*** surgery/chemotherapy/radiation/trastuzumab-era indicator
- ***Temporal/region:*** era/diagnosis year, SEER registry region

Potential Effect Modifiers:

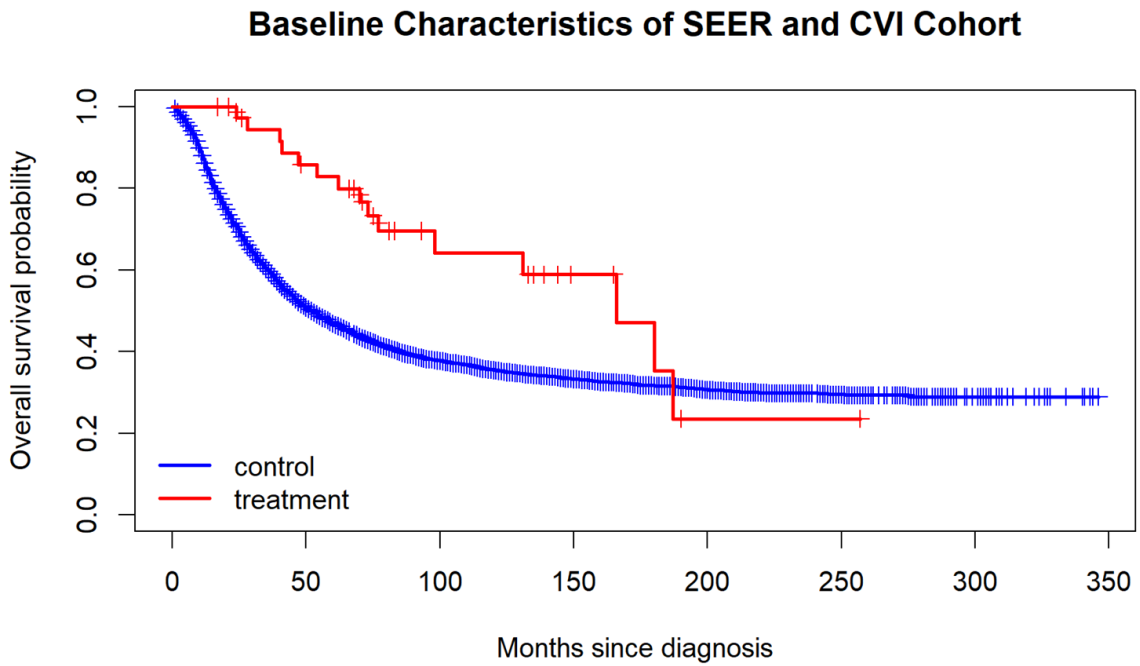
- *HR status, age, stage, era.*

Matching Strategy

- 1:k nearest neighbor matching (e.g., k = 1–5) with a caliper of 0.2 standard deviations of the propensity score.
- P(treatment)=f(AgeBin,Stage,HR status,DxYear)
- Balance check: standardized mean difference (SMD) < 0.1.

$$\mathrm{SMD} = \frac{\mathrm{Mean}_{\mathrm{treated}} - \mathrm{Mean}_{\mathrm{control}}}{\mathrm{SD}_{\mathrm{pooled}}}$$

Data Exploration



Characteristic	control N = 3,875[1]	treatment N = 40[1]
ER		
neg	1,939 (50%)	26 (67%)
pos	1,936 (50%)	13 (33%)
Unknown	0	1
PR		
neg	2,297 (59%)	23 (62%)
pos	1,578 (41%)	14 (38%)
Unknown	0	3
TrastuzumabUse	1,199 (31%)	25 (63%)
Dead		
FALSE	1,461 (38%)	25 (63%)
TRUE	2,414 (62%)	15 (38%)
Survival months	42 (18, 113)	76 (48, 134)
Stage		
IIIB	2,437 (63%)	30 (75%)
IIIC	902 (23%)	1 (2.5%)
IV	536 (14%)	9 (23%)
Age at Diagnosis	55 (45, 65)	45 (35, 53)

[1] n (%); Median (Q1, Q3)

After Matching

Sub sample	SEER Control	N	CVI Treatment	N
SS1	No trastuzumab use (DxYear < 2006)	60	No trastuzumab use	12
SS2	Trastuzumab-era (DxYear > 2010 but no HER2 restriction)	113	With trastuzumab use	25
Total (matched)		173		37

Characteristic	control N = 173[1]	treatment N = 37[1]
Age at Diagnosis	55 (45, 65)	55 (45, 65)
ER		
neg	111 (64%)	25 (68%)
pos	62 (36%)	12 (32%)
Unknown	0 (0%)	0 (0%)
PR		
neg	121 (70%)	23 (62%)
pos	52 (30%)	14 (38%)
Unknown	0 (0%)	0 (0%)
Stage		
IIIB	150 (87%)	28 (76%)
IIIC	5 (3%)	1 (3%)
IV	18 (10%)	8 (22%)
Dead		
FALSE	89 (51%)	25 (68%)
TRUE	84 (49%)	12 (32%)
Survival months	57 (25, 96)	77 (54, 135)

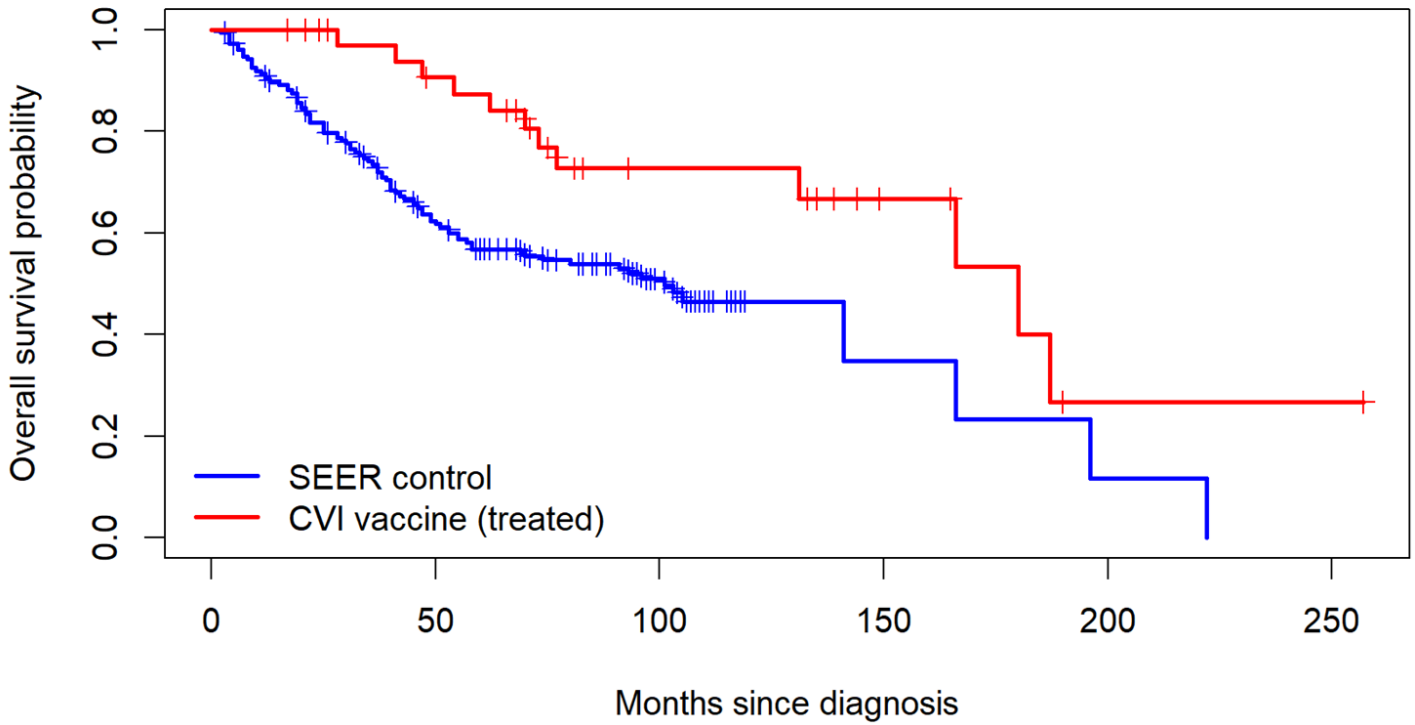
[1] n (%); Median (Q1, Q3)

Primary Analysis

Group	N	Events	Median OS (months, 95% CI)
SEER control	173	83	101 (58–NA)
CVI vaccine	37	12	180 (166–NA)

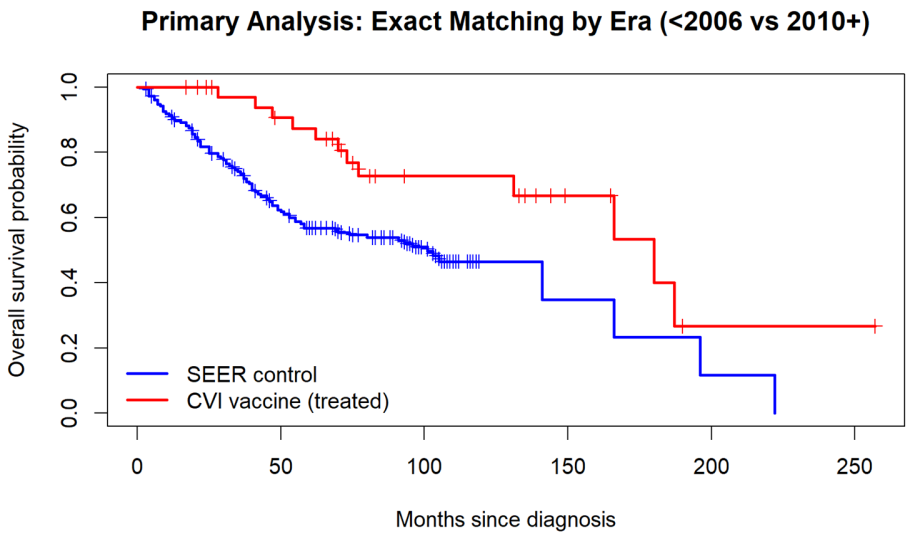
Median overall survival estimated using weighted Kaplan–Meier curves after propensity score matching.
NA indicates that the upper confidence bound for the median was not estimable due to limited follow-up.

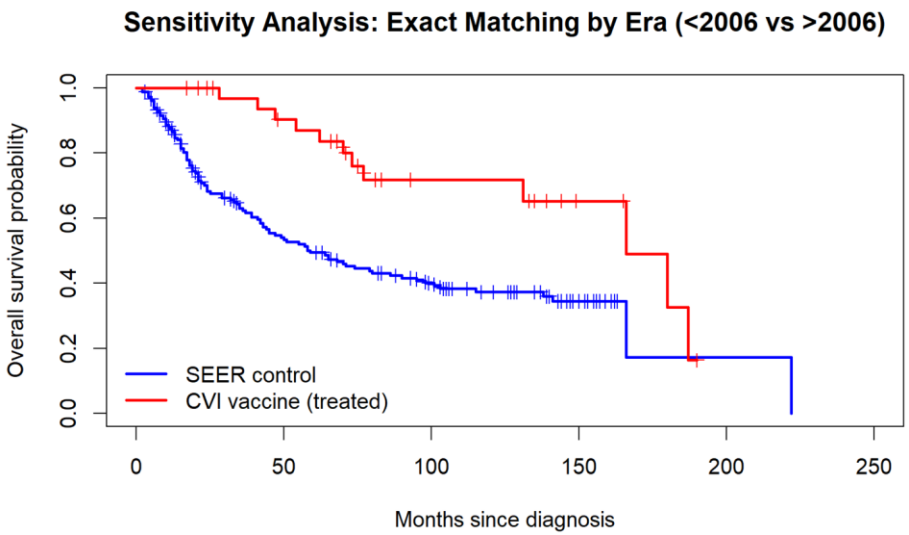
Primary Analysis: Exact Matching by Era (<2006 vs 2010+)



Proportional hazards assumption evaluated using scaled Schoenfeld residuals ($p > 0.05$).

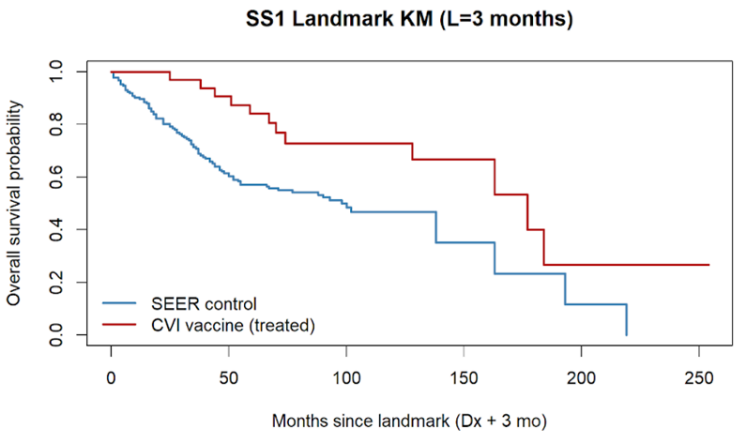
Sensitivity Analysis

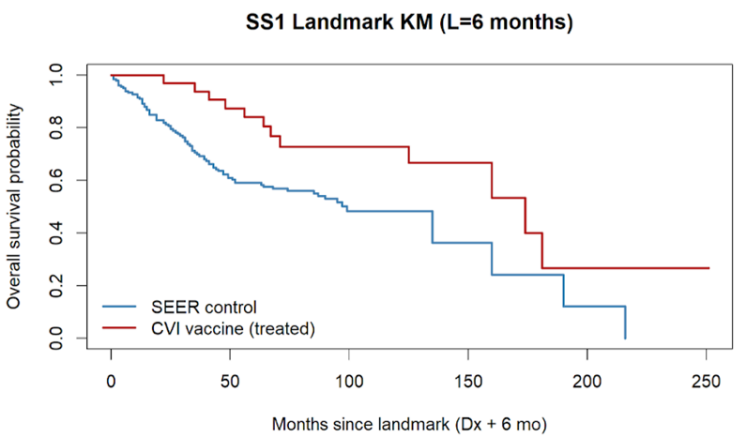


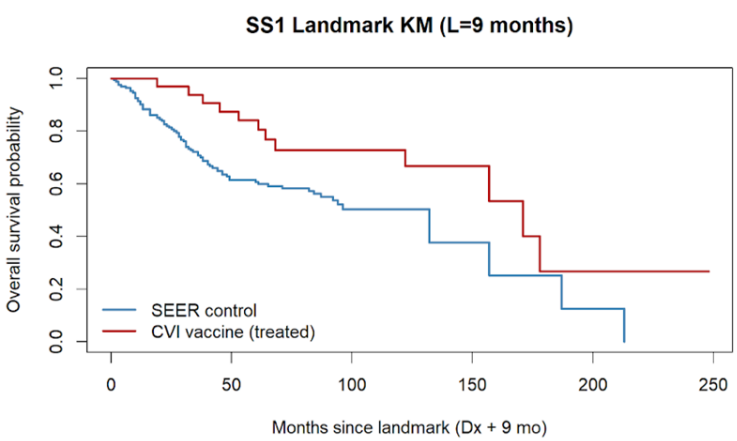


	Primary	Sensitivity
Matching Era	<2006 vs 2010+	<2006 vs >2006
Matched N (Control / Treated)	173 / 37	177 / 36
HR (95% CI)	0.42 (0.26–0.67)	0.37 (0.24–0.57)
p-value	0.0003	<0.001

Landmark Analysis (3, 6, 9 months)







Landmark (mo)	N (Control/ Treated)	HR (95% CI)	p-value
3	171 / 37	0.42 (0.27–0.68)	0.0003
6	163 / 37	0.45 (0.28–0.72)	0.0008
9	157 / 37	0.47 (0.30–0.76)	0.0021

Secondary Analysis

Table 2: Effect modification of treatment effect using landmark analysis at 6 months

Modifier	Stratum	N control	N treated	Events control	Events treated	HR (95% CI)	p-value	p-interaction
HR status	HR−	93	23	49	8	0.32 (0.17–0.60)	0.0004	0.236
	HR+	70	14	28	4	0.69 (0.33–1.44)	0.321	
Age group	<50	104	23	54	8	0.47 (0.26–0.83)	0.010	0.829
	≥50	59	14	23	4	0.43 (0.17–1.07)	0.069	
Stage	III	147	29	69	9	0.48 (0.29–0.82)	0.0069	0.336
	IV	16	8	8	3	0.32 (0.13–0.79)	0.0137	

HR: hazard ratio estimated from weighted Cox proportional hazards models with landmark time at 6 months.
p-values correspond to within-stratum treatment effects.
p-interaction values are from Wald tests of treatment-by-modifier interaction terms.

Conclusion

- Patients receiving the HER2-directed vaccine demonstrated significantly improved overall survival compared with matched SEER controls.
- The weighted Cox proportional hazards model estimated a 58% lower hazard of death in the HER2-directed vaccine group (HR = 0.42, 95% CI 0.26–0.67; $p < 0.001$).
- The survival benefit of treatment was generally consistent across most subgroups, with no statistically significant evidence for effect modification.
- Sensitivity analyses using an alternative era definition yielded similar treatment effects, supporting the robustness of primary findings.

Limitations

- Trastuzumab use was not directly measured in both data sources, and therefore may result in exposure misclassification and residual confounding.
- HER2 information only recorded after 2010 in SEER database.
- Stage IIIB, IV, and dxyear has a SMD > 0.1, indicating some residual imbalance remaining, which may introduce residual confounding.

Variables	SMD
Stage IIIB	0.265
Stage IV	0.276
DxYear	0.890

Reference

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